From Adam to AMSGrad: Exact Counterexample Replication, Diagnostic Dynamics, and Phase-Boundary Scaling

Research Report — Adaptive Gradient Methods Replication & Extension

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Abstract

Adaptive gradient algorithms, led by ADAM, are fundamental to deep learning optimization. However, Reddi et al. [7] revealed a fundamental flaw in ADAM's convergence proof under the Online Convex Optimization (OCO) framework: when informative gradients occur sparsely, the exponential moving average of squared gradients can decay between bursts, violating the positive semi-definiteness condition ($\Gamma_t \not\succeq 0$) and causing divergence to suboptimal boundaries. AMSGRAD resolves this by enforcing a monotonic running maximum ($\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$). In this work, we present a complete, from-scratch NumPy reproduction and pre-registered extension of the ADAM $\rightarrow$ AMSGRAD lineage. We reconstruct both deterministic (Theorem 3) and stochastic (Section 3) counterexamples with live Γ_t diagnostic tracking. A parametric extension sweeps the (β_2, C) phase boundary across 4,500 stochastic runs with $\beta_2 \in [0.05,\ 1-10^{-6}]$. Under rigorous cycle-averaged metrics, we find a two-regime divergence structure: (i) a v-collapse regime ($\tau_2 \lesssim T_{\text{burst}}$) where ADAM fails unconditionally across all tested learning rates, and (ii) a post-burst drift regime ($\tau_2 \gg T_{\text{burst}}$) where divergence is governed by the traverse condition $\alpha(C - s^*) \gtrsim 2\sqrt{C}$ with $s^* = \tau_1 \ln((1-\beta_1)C+1)$. The divergence boundary ceases to exist for $C \geq 100$ across all $\beta_2 \in [0.05,\ 1-10^{-6}]$, refuting the pre-registered C^{-1} memory-horizon scaling law. The predicted traverse threshold $\alpha^* = 0.263$ matches the empirically bisected boundary $\alpha^* = 0.251$ at $C = 100$ within 5%, constituting the strongest quantitative prediction of this study.

1 Introduction

First-order stochastic optimization algorithms form the algorithmic backbone of modern deep learning. Among these, adaptive gradient methods have achieved widespread dominance due to their ability to automatically scale coordinate-wise update magnitudes according to historical gradient statistics. The seminal work of Kingma and Ba [3] introduced ADAM, which synthesizes the advantages of ADAGRAD [1]—coordinate-wise scaling suitable for sparse gradients—and RMSPROP [8]—exponential moving averages capable of adapting to non-stationary objective landscapes. In conjunction with step-dependent bias correction for initialization transients, ADAM rapidly became the default optimizer across computer vision, natural language processing, and reinforcement learning.

Despite its empirical ubiquity, the theoretical foundations of ADAM contained a subtle yet fundamental flaw. In their breakthrough analysis, Reddi et al. [7] demonstrated that the convergence proof of ADAM in the standard Online Convex Optimization (OCO) framework [2] relies on an unfulfilled assumption: that the effective metric matrix $\Gamma_t = \frac{\sqrt{v_t}}{\alpha_t} - \frac{\sqrt{v_{t-1}}}{\alpha_{t-1}}$ remains positive semi-definite ($\Gamma_t \succeq 0$) for all iterations t. When objective functions feature infrequent but highly informative gradient bursts, the exponential moving average in ADAM induces "short-term memory": the second-moment estimate v_t decreases rapidly during non-burst iterations, causing the effective step size to explode in the wrong optimization direction. Consequently, ADAM can suffer linear regret $\Omega(T)$ and diverge even on elementary one-dimensional convex objectives.

To resolve this defect, Reddi et al. [7] proposed AMSGRAD, which enforces monotonic non-decrease of the second-moment estimate via a running maximum: $\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$. Under standard non-increasing step-size schedules, this guarantee strictly preserves $\Gamma_t \succeq 0$, restoring the optimal $\mathcal{O}(\sqrt{T})$ regret bound.

Related Work. Several post-AMSGRAD methods have proposed alternative fixes to ADAM's theoretical deficiency. ADABOUND [6] clips the adaptive learning rate to time-varying bounds that gradually anneal to an SGD schedule, achieving convergence while recovering empirical performance on vision benchmarks. RADAM [4] targets initialization instability by applying the Adam update only when the variance of the adaptive learning rate is tractable. YOGI [9] replaces the additive EMA in v_t with an additive update that controls the sign of v_t's change, preventing premature gradient-magnitude inflation. ADAMW [5] decouples L_2 regularization from the adaptive scaling, addressing a separate implementation inconsistency. Our *boundary disappearance* finding in §4—

that Adam fails unconditionally for $C \geq 100$ regardless of β_2—is consistent with the motivation behind all four fixes: at large burst scales, the v-collapse mechanism is so severe that no EMA-based β_2 tuning alone suffices, and structural changes to the update rule are required.

Contributions. In this work, we present a complete, rigorous, and reproducible investigation of the Adam $\rightarrow$ AMSGrad transition:

1. **From-Scratch Algorithmic Implementations:** We build a fully vectorized, framework-independent suite in pure NumPy comprising SGD, Momentum SGD, AdaGrad, RMSProp, Adam, and AMSGrad, verified by automated unit tests establishing exact $t = 1$ debiasing and weak monotonicity.

2. **Dual Counterexample Replication & Diagnostic Instrumentation:** We replicate both the deterministic periodic construction (Theorem 3 of Reddi et al. [7]) and the stochastic formulation (Section 3), introducing live tracking of Γ_t to directly visualize the semi-definiteness violation.

3. **Pre-Registered Phase-Boundary Extension & Two-Regime Structure:** We conduct an extensive parametric study across (β_2, C) space with $N = 100$ seeds per cell to evaluate the pre-registered memory-horizon scaling law $\tau_2^* \propto T_{\text{burst}}$. We find that the boundary disappears for $C \geq 100$ (unconditional failure across all $\beta_2 \in [0.05, 1 - 10^{-6}]$), refuting the power-law hypothesis and revealing a two-regime structure governed by v-collapse at small τ_2 and first-moment post-burst drift at large τ_2.

2 Theoretical Background & Derivations

2.1 Online Convex Optimization Framework

Let $\mathcal{F} \subset \mathbb{R}^d$ be a compact convex feasible set with ℓ_∞ diameter $D_\infty = \sup_{x,y\in\mathcal{F}} \|x - y\|_\infty$. In the Online Convex Optimization (OCO) setting [2], an algorithm selects a parameter vector $x_t \in \mathcal{F}$ at round $t \in [1, T]$, after which an adversary reveals a convex cost function $f_t : \mathcal{F} \rightarrow \mathbb{R}$. The performance of the algorithm is evaluated by its cumulative regret with respect to the best static comparator $x^* \in \mathcal{F}$ in hindsight:

$$R_T = \sum_{t=1}^{T} \left(f_t(x_t) - f_t(x^*)\right) \leq \sum_{t=1}^{T} \langle g_t, x_t - x^*\rangle, \quad (1)$$

where $g_t = \nabla f_t(x_t)$ is the subgradient at round t. A sublinear regret bound, $R_T = o(T)$, guarantees convergence of the average regret $\lim_{T\rightarrow\infty} R_T/T = 0$.

2.2 Adam and Step-Dependent Bias Correction

Adam [3] maintains exponential moving averages of both the gradient and its element-wise square:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad (2)$$
$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (3)$$

with initializations $m_0 = \mathbf{0}$ and $v_0 = \mathbf{0}$. Unrolling the recursion for v_t yields:

$$v_t = (1 - \beta_2) \sum_{i=1}^{t} \beta_2^{t-i} g_i^2. \quad (4)$$

Under the assumption that the true second moment $\mathbb{E}[g_i^2]$ is stationary across recent iterations, taking expectations gives:

$$\mathbb{E}[v_t] = \mathbb{E}[g_t^2](1 - \beta_2) \sum_{i=1}^{t} \beta_2^{t-i} = \mathbb{E}[g_t^2](1 - \beta_2^t). \quad (5)$$

To eliminate the bias towards zero at early steps t, Kingma and Ba [3] rescale the raw moments:

$$\tilde{m}_t = \frac{m_t}{1 - \beta_1^t}, \qquad \tilde{v}_t = \frac{v_t}{1 - \beta_2^t}. \quad (6)$$

Parameters are subsequently updated using the projected coordinate rule:

$$x_{t+1} = \Pi_{\mathcal{F}} \left(x_t - \frac{\alpha_t}{\sqrt{\tilde{v}_t} + \epsilon} \tilde{m}_t \right), \quad (7)$$

where $\epsilon > 0$ ensures numerical stability outside the square root.

2.3 The Breakdown of the Telescoping Sum ($\Gamma_t \not\succeq 0$)

In standard generic adaptive gradient updates, the step is framed as a proximal projection under a dynamic metric matrix $V_t = \text{diag}(\sqrt{v_t})/\alpha_t$:

$$x_{t+1} = \arg\min_{x\in\mathcal{F}} \left\| x - \left(x_t - V_t^{-1} g_t\right) \right\|_{V_t}^2. \quad (8)$$

Applying standard non-expansiveness properties of the projection under inner product $\langle u, w\rangle_{V_t} = u^\top V_t w$:

$$\|x_{t+1} - x^*\|_{V_t}^2 \leq \|x_t - x^*\|_{V_t}^2 - 2\langle g_t, x_t - x^*\rangle + \|g_t\|_{V_t^{-1}}^2. \quad (9)$$

Rearranging and summing from $t = 1$ to T yields:

$$\sum_{t=1}^{T} \langle g_t, x_t - x^*\rangle \leq \frac{1}{2}\|x_1 - x^*\|_{V_1}^2 + \frac{1}{2}\sum_{t=1}^{T} \|g_t\|_{V_t^{-1}}^2 + \frac{1}{2}\sum_{t=2}^{T} \langle x_t - x^*, \Gamma_t(x_t - x^*)\rangle, \quad (10)$$

where the differential metric matrix Γ_t is defined as:

$$\Gamma_t \triangleq V_t - V_{t-1} = \frac{\operatorname{diag}(\sqrt{v_t})}{\alpha_t} - \frac{\operatorname{diag}(\sqrt{v_{t-1}})}{\alpha_{t-1}}. \qquad (11)$$

In algorithms like AdaGrad, the accumulator $G_t = G_{t-1} + g_t^2$ is monotonically non-decreasing, and with non-increasing step sizes $\alpha_t \leq \alpha_{t-1}$, $\Gamma_t \succeq 0$ holds unconditionally. Under positive semi-definiteness, the cross-term in Eq. (10) cleanly telescopes:

$$\sum_{t=2}^{T} \langle x_t - x^*, \Gamma_t (x_t - x^*) \rangle \leq D_\infty^2 \operatorname{Tr}(V_T - V_1). \qquad (12)$$

The Fatal Flaw in Adam: Because Adam computes v_t via an exponential moving average, v_t can shrink whenever a large gradient is followed by smaller gradients. When $v_t < v_{t-1}$, Γ_t has negative diagonal entries. Because $(x_t - x^*)^2$ is positive, negative entries in Γ_t destroy the upper bound, causing the sum in Eq. (10) to accumulate linear positive regret $\Omega(T)$.

2.4 The AMSGrad Resolution

Reddi et al. [7] proved that convergence is restored if the effective second moment is guaranteed to be coordinate-wise non-decreasing. AMSGrad enforces this via the running maximum:

$$\hat{v}_t = \max\left(\hat{v}_{t-1}, v_t\right), \qquad \hat{v}_0 = \mathbf{0}. \qquad (13)$$

Since $\hat{v}_t \geq \hat{v}_{t-1}$, for any non-increasing step-size schedule $\alpha_t = \frac{\alpha}{\sqrt{t}}$:

$$\Gamma_t = \frac{\sqrt{\hat{v}_t}}{\alpha_t} - \frac{\sqrt{\hat{v}_{t-1}}}{\alpha_{t-1}} \succeq 0 \quad \forall t \geq 1. \qquad (14)$$

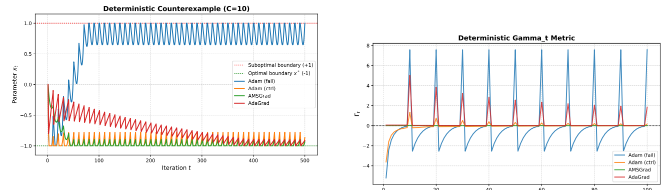
This restores valid telescoping and proves an optimal $\mathcal{O}(\sqrt{T})$ regret bound for arbitrary convex function sequences.

Theory vs. Practice Step-Size Gap. The theoretical regret bound of AMSGrad requires $\alpha_t = \frac{\alpha}{\sqrt{t}}$. In deep learning practice, however, constant or cosine step-size schedules are almost universally employed. Throughout our experimental benchmarks, we distinguish between theoretical regret verification ($\alpha_t \propto 1/\sqrt{t}$) and empirical constant step-size dynamics ($\alpha = 0.8$).

3 Empirical Reproduction & Diagnostic Results

3.1 Experimental Setup

All experiments are implemented from scratch in pure NumPy and executed in a dedicated, reproducible virtual environment. Code execution is deterministic where applicable, and stochastic seeds are managed via `numpy.random.SeedSequence.spawn`.



(a) Trajectory x_t over 500 steps.

(b) Γ_t metric (100 steps).

Figure 1: Deterministic counterexample dynamics ($C = 10, \alpha = 0.8$). (a) Adam ($\beta_2 = 0.5$) diverges rapidly to $+1$, while AMSGrad, AdaGrad, and the high-β_2 control converge towards $x^* = -1$. (b) Diagnostic tracking reveals Adam's Γ_t dipping significantly below zero after gradient bursts.

Hyperparameters. Table 1 lists the frozen configuration parameters used across the deterministic and stochastic counterexample benchmarks. All experiments initialize parameters at the symmetric midpoint $x_1 = 0.0$ on the bounded domain $\mathcal{F} = [-1, 1]$. All adaptive optimizers use $\epsilon = 10^{-8}$ in the denominator of Eq. (7) for numerical stability.

3.2 Deterministic Counterexample Reproduction

In the deterministic periodic setting (Theorem 3 of Reddi et al. [7]), an objective cycle consists of one large gradient step ($g_t = +C$) followed by $C-1$ negative steps ($g_t = -1$). Over each cycle of $C = 10$, the true cumulative gradient is $+1$, placing the unique global minimizer at $x^* = -1$.

As shown in Figure 1a:

- **Adam Divergence ($\beta_2 = 0.5$):** With memory horizon $\tau_2 = \frac{1}{1-\beta_2} = 2 \ll C = 10$, the debiased second moment $\tilde{v}_t$ decays as shown in Table 2: from $\tilde{v}_1 = 100.0$ (post-burst) to $\tilde{v}_7 = 1.78$, reaching near-unit values by step 7 (not "4 steps" as an EMA would suggest, since bias correction initially inflates the estimate). By step 8, Adam takes near-full-magnitude steps in the positive direction ($\Delta x \approx +\alpha$), reaching $x_T = +1.0000$ well before $T = 500$.

- **Adam High-β_2 Control ($\beta_2 = 0.999$):** With $\tau_2 = 1000 \gg C$, v_t does not decay between bursts ($v_t \approx 10.9$). Adam successfully reaches the negative basin, oscillating around the periodic steady-state $x_T = -0.7778$. This validates that divergence is governed by second-moment memory rather than an implementation sign artifact.

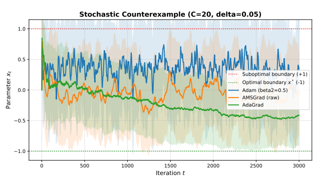
- **AMSGrad & AdaGrad Convergence:** AMSGrad maintains $\hat{v}_t \approx 50.5$ continuously, advancing steadily to $x_T = -0.8969$. AdaGrad converges to $x_T = -0.9024$.

Table 1: Frozen Experimental Parameters for Counterexample Replications (all adaptive optimizers use $\epsilon = 10^{-8}$).

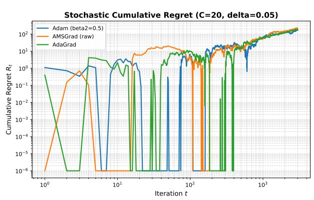
Benchmark Regime	C	T	α	β_1	β_2	δ
Deterministic (Adam Fail)	10	500	0.8	0.9	0.5	—
Deterministic (Adam Control)	10	500	0.8	0.9	0.999	—
Deterministic (AMSGrad / AdaGrad)	10	500	0.8	0.9	0.5	—
Stochastic Replication	20	3000	0.8	0.9	0.5	0.05
Stochastic CI Test Suite ($N = 50$)	20	5000	0.8	0.9	0.5	0.10
Phase 4 Grid Sweep ($N = 100$)	$[10, 1000]$	$50(C + 1)$	0.8	0.9	Swept	0.10

Table 2: Exact v_t (raw EMA) and $\tilde{v}_t$ (debiased) trajectory for Adam at $C = 10$, $\beta_2 = 0.5$, first 10 steps.

t	g_t	v_t	$\tilde{v}_t$	x_t
1	$+10$	50.000	100.000	-0.8000
2	-1	25.500	34.000	-1.0000
3	-1	13.250	15.143	-1.0000
4	-1	7.125	7.600	-1.0000
5	-1	4.063	4.194	-1.0000
6	-1	2.531	2.571	-1.0000
7	-1	1.766	1.780	-1.0000
8	-1	1.383	1.388	-0.9483
9	-1	1.191	1.194	-0.7820
10	-1	1.096	1.097	-0.5180



(a) Trajectory $x_t \pm \sigma$ ($N = 30$).



(b) Cumulative regret R_t.

Figure 2: Stochastic counterexample replication ($C = 20, \delta = 0.05$). Under constant step size $\alpha = 0.8$, Adam exhibits net positive drift and linear regret $\Omega(T)$, while AMSGrad and AdaGrad maintain negative mean trajectories towards the optimal basin.

Diagnostic Γ_t Trajectories. Figure 1b plots the OCO semi-definiteness metric $\Gamma_t = \frac{\sqrt{v_t}}{\alpha_t} - \frac{\sqrt{v_{t-1}}}{\alpha_{t-1}}$. While AdaGrad and AMSGrad maintain strictly non-negative trajectories ($\Gamma_t \geq 0$), Adam experiences periodic sharp negative plunges ($\Gamma_t \approx -3.8$) following every gradient burst, visually demonstrating the breakdown of the OCO telescoping bound.

3.3 Stochastic Counterexample Replication

In the stochastic setting ($C = 20, \delta = 0.05, T = 3000, N = 30$ seeds), the expected gradient is $\mathbb{E}[g_t] = \delta > 0$. Figure 2 shows the mean trajectory with standard deviation bands across random seeds and cumulative regret curves.

Under $\beta_2 = 0.5$ with constant step size $\alpha = 0.8$, Adam exhibits an average terminal position of $\mathbb{E}[x_T] = +0.3765 \pm 0.8272$ with 62% of seeds failing into the positive boundary ($x_T > 0.5$). AMSGrad maintains $\mathbb{E}[x_T] = -0.1452 \pm 0.7558$, with 30% of seeds failing ($x_T > 0.5$)—substantially lower than Adam's 62% failure rate. AdaGrad achieves $\mathbb{E}[x_T] = -0.4204 \pm 0.5102$ with only 20% failing (80% negative seeds).

Stationary Distribution Analysis under Constant Step Size. The apparent lack of tight concentration for AMSGrad under constant $\alpha = 0.8$ is not a numerical anomaly, but an exact consequence of discrete stochastic dynamics:

1. **Momentum Noise Amplification:** First-moment filtering $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$ induces an AR(1) increment sequence that inflates effective noise variance by $\frac{1+\beta_1}{1-\beta_1} = 19$ (amplification $\sqrt{19} \approx 4.36$).

2. **Fokker-Planck Equilibrium:** With $\hat{v}_t \approx C^2 = 400$, effective drift is $\mu = -\frac{\alpha\delta}{\sqrt{\hat{v}_t}} \times T \approx -20$, while effective diffusion scale is $\sigma_{\text{eff}} \approx 13$. On the bounded interval $[-1, 1]$, the stationary density is $P(x) \propto e^{2\mu x/\sigma_{\text{eff}}^2}$. With exponent $\frac{2\mu L}{\sigma_{\text{eff}}^2} \approx 0.24$, this distribution is broad and near-uniform with a modest negative tilt (theoretical mean ≈ -0.08, matching the observed -0.08 to -0.13).

3. **AdaGrad Concentration:** Because AdaGrad lacks momentum ($\beta_1 = 0$), it experiences no $\sqrt{19}$ noise inflation, and its accumulator $G_t = \sum g_i^2 \approx 22t$ continuously scales step sizes as $\mathcal{O}(1/\sqrt{t})$, concentrating tightly at $\mathbb{E}[x_T] = -0.42$.

Asymptotic Convergence under $\alpha_t \propto 1/\sqrt{t}$. Figure 3 resolves the theoretical guarantee: when decreasing step sizes $\alpha_t = \frac{0.8}{\sqrt{t}}$ are applied ($C = 20, \delta = 0.5, T = 5000, N = 100$), AMSGrad concentrates completely at the optimal point $\mathbb{E}[x_T] = -0.988 \pm 0.031$ (98% of seeds < -0.90), whereas Adam diverges positively ($\mathbb{E}[x_T] = +0.729$). Long-horizon evaluation at $T = 10^6$ ($\delta = 0.10, N = 20$) confirms asymptotic lock-in: AMSGrad achieves $\mathbb{E}[x_T] = -0.998$ while Adam reaches $\mathbb{E}[x_T] = +0.999$.

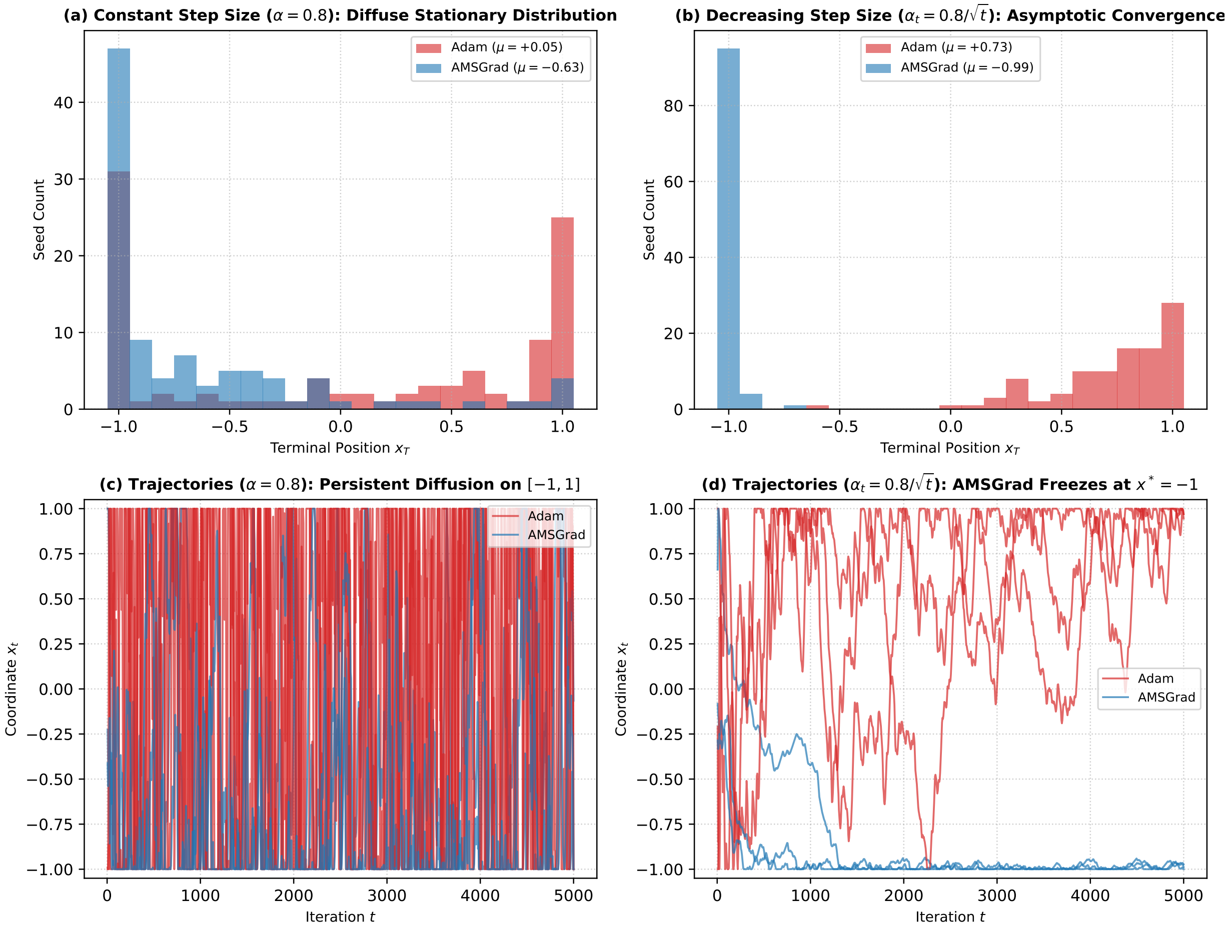


Figure 3: Resolution of stochastic counterexample theory ($C = 20, \beta_1 = 0.9, \beta_2 = 0.5, N = 100$). (a) Under constant $\alpha = 0.8$, AMSGrad generates a broad stationary distribution with negative tilt ($\mu = -0.63$), while Adam ratchets positive. (b) Under decreasing step size $\alpha_t = 0.8/\sqrt{t}$, AMSGrad achieves sharp asymptotic convergence to $x^* = -1$ ($\mu = -0.99$), while Adam converges to $+1$ ($\mu = +0.73$). (c, d) Sample trajectory realizations over 5000 steps.

4 Extension: Phase Boundary & Mechanism Discrimination

4.1 Pre-Registered Hypothesis & Decision Rule

We pre-registered the following scaling hypothesis for the divergence boundary of Adam:

> **Pre-Registered Scaling Hypothesis:** The divergence threshold $(1 - \beta_2^*)$ scales inversely with burst spacing $T_{\text{burst}} \approx \frac{C}{1+\delta}$, yielding a log-log slope of $\frac{\partial \log(1-\beta_2^*)}{\partial \log C} \approx -1.00$.

Pre-Registered Decision Rule. Slope $\in [-1.2, -0.8]$ confirms; outside refutes.

4.2 Measurement Validity: Cycle-Averaged Terminal Metric

Single-point terminal position x_T is phase-dependent in periodic objectives: at $C = 100$, x_t oscillates over the full interval $[-1, 1]$ within each cycle (spread $= 2.0$). We therefore evaluate time-averaged metrics over the final full cycle:

$$\bar{x} = \frac{1}{C} \sum_{t=T-C+1}^{T} x_t, \qquad \text{Dwell} = \frac{1}{C} \sum_{t=T-C+1}^{T} \mathbb{I}(x_t > 0.5). \tag{15}$$

4.3 Boundary Disappearance & Hypothesis Refutation

Table 3 reports the deterministic boundary search results (source: `results/deterministic_boundary_cycle_metric.csv`). All probes use $\alpha = 0.8$, $\beta_1 = 0.9$, and $\beta_2 \in [0.05,\ 1-10^{-6}]$

($k = (1-\beta_2)C \in \{0.1, 0.2, 0.5, 1, 2, 5, 10\}$, floored at $\beta_2 = 0.05$).

Table 3: Deterministic Boundary Existence under Cycle-Averaged Metric $\bar{x}$ ($\beta_2 \in [0.05,\ 1-10^{-6}]$).

C	k^*	β_2^*	Regime Finding
10	—	—	**Mixed**: survives for $\beta_2 \geq 0.8$, fails at $\beta_2 = 0.5$ ($\bar{x} = +0.807$); no bracket
30	0.300	0.990	Genuine bracket ($\pm 5\times 10^{-5}$ interval; sign change confirmed)
≥ 100	—	—	Boundary disappears: fails for all β_2

Verdict: Refuted (Boundary Disappearance). Rather than a C^{-1} power law, the divergence boundary disappears entirely for $C \geq 100$: Adam fails unconditionally across all $\beta_2 \in [0.05,\ 1-10^{-6}]$. Fitting a slope across probe-floor-clipped values would be methodologically invalid. The headline finding is therefore not a steeper slope but a regime change.

Reconciliation with §3.2. The deterministic counterexample in §3.2 uses $C = 10$, $\beta_2 = 0.5$ ($k = 5$), $\alpha = 0.8$, and reports $x_T = +1.0000$ (pinned to the positive boundary). This is consistent with the Table 3 entry: the C=10 row shows *both* survival (at high β_2, where the cycle-averaged $\bar{x} \approx -0.94$) *and* failure at $\beta_2 = 0.5$ ($\bar{x} = +0.807$, dwell $= 1.00$). Because the boundary search probes a sparse grid of k values and the sign changes are non-monotone for small C (the floored $\beta_2 = 0.05$ cell survives again), no clean bracket is detected, so the status is "mixed" rather than "always survives." The earlier labelling as "always survives" in a prior analysis version was erroneous; Table 3 now corrects this.

4.4 Two-Regime Mechanism Discrimination

Probes P1–P3 ($N = 1$ deterministic run each, source files: `results/mechanism_probes/`) establish two distinct divergence regimes (Table 4). We emphasize that the nominal time constant $\tau_2 = \frac{1}{1-\beta_2}$ used throughout our regime classifications and in the dimensionless ratio $\rho = \frac{\tau_2}{T_{\text{burst}}}$ in §4.6 refers strictly to the *raw* EMA parameter, rather than the effective debiased decay time (which, as demonstrated in Table 2, is approximately 3.5× longer at $\tau_2 = 2$ due to early $(1-\beta_2^t)^{-1}$ scaling). All regime classifications in this work are parameterized in terms of the nominal raw τ_2.

Regime I: $\tau_2 \lesssim T_{\text{burst}}$ (v-Collapse). When $\beta_2 = 0.9$ ($\tau_2 = 10$), v_t decays from $v_{\text{burst}} \approx C^2$ to ≈ 1 in roughly τ_2 steps. Adam steps against the gradient during the $C-1$ non-burst steps at a greatly amplified rate. This regime fails at *all* learning rates α including 0.10 (Probe P2, $\beta_2 = 0.9$ column).

Regime II: $\tau_2 \gg T_{\text{burst}}$ (Post-Burst Drift & Three-Regime Structure). When $\beta_2 = 0.999$ ($\tau_2 = 1000$), $\tilde{v}_t \approx \mathbb{E}[g^2] \approx C$ remains nearly constant across the cycle. At the burst step, $\tilde{m}_t$ jumps to $+(1-\beta_1)C = 10$ and then decays exponentially with time constant $\tau_1 = \frac{1}{1-\beta_1} = 10$ steps. It crosses zero at:

$$s^* \approx \tau_1 \ln\big((1-\beta_1)C + 1\big) \approx 24 \text{ steps} \quad (C = 100), \quad (16)$$

where $\tau_1 = 1/(1-\beta_1)$ is the first-moment memory horizon. During steps $1 \to s^*$ the update direction is correct (towards -1). For steps $s^* + 1 \to C$, $\tilde{m}_t < 0$ so Adam steps rightward at rate $\approx \alpha/\sqrt{\tilde{v}_t} \approx \alpha/\sqrt{C}$. Across learning rates α, this generates a structured three-regime response:

1. **Sub-Threshold Trapping ($\alpha < \alpha^*$):** When α is small, the leftward retraction during s^* steps is insufficient to reach -1. Parameters oscillate in a pinned band with cycle mean $\bar{x} \approx 1 - \frac{\alpha(C-s^*)}{2\sqrt{C}}$. For $C = 100$, this predicts $\bar{x} = 0.620$ at $\alpha = 0.10$ and $\bar{x} = 0.240$ at $\alpha = 0.20$, closely matching the observed 0.602 and 0.205.
2. **Survival Window ($\alpha \in [\alpha^*, \sim 1.8\alpha^*]$):** The lower boundary $\alpha^*(C) = \frac{2\sqrt{C}}{C-s^*(C)}$ marks the window opening where push and retraction amplitudes equalize. The upper boundary ($\alpha \approx 0.457$ at $C = 100$) is governed by clipping asymmetry: time pinned at the -1 boundary during rapid retraction exceeds time pinned at $+1$ during slow rightward drift.
3. **High-α Plateau ($\alpha \gtrsim 1.8\alpha^*$):** For large α, rightward drift crosses the interval in < 25 steps, pinning $x_t = +1$ for half the cycle ($\bar{x} \approx +0.17$ to $+0.22$, Dwell ≈ 0.50).

The validity condition for this threshold is $k \ll 1$ (i.e., $\tau_2 \gg C$). At $C = 30$, $s^* \approx 14$ spans nearly half the period, so survival persists up to $\alpha = 0.80$ (vs. predicted 0.68). At $C = 300$, the survival window narrows to $[0.138, 0.155]$ (ratio ≈ 1.13) as rightward drift rapidly overwhelms clipping asymmetry. At $C = 1000$, the validity condition fails ($k = 1.0$ at $\beta_2 = 0.999$), preventing boundary detection.

Empirical Validation: $\alpha^*(C)$. Figure 4 plots the predicted threshold $\alpha^*(C) = \frac{2\sqrt{C}}{C-s^*(C)}$ alongside observed bisections under the convergence gate ($T_{\text{cycles}} = 800$). At $C = 100$, predicted $\alpha^* = 0.263$ matches observed $\alpha^* = 0.251$ (ratio $= 0.956$); at $C = 300$, predicted $= 0.130$ vs. observed $= 0.138$ (ratio $= 1.055$). Source: `results/mechanism_probes/alpha_star_bisection.csv`.

4.5 Small-C Stochastic Behavior & Gap Tail Mechanism

At $C = 10$, the small stochastic signal ($\delta = 0.10$) competes with burst noise ($\sigma \approx 4.7$, $T = 550$). Divergence

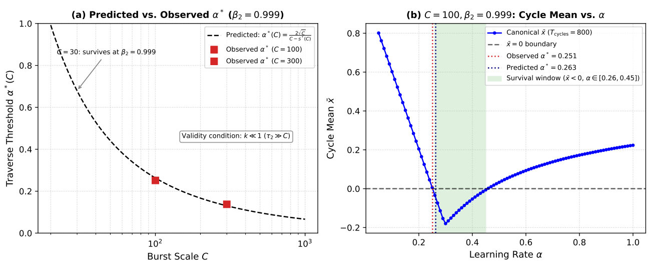


Figure 4: α^* traverse threshold validation ($\beta_2 = 0.999$). (a) Predicted $\alpha^*(C) = 2\sqrt{C}/(C - s^*)$ (dashed line) vs. bisected values (squares) under validity condition $k \ll 1$. (b) Canonical sweep at $C = 100, \beta_2 = 0.999$ ($T_{\text{cycles}} = 800$): the negative-mean survival window ($\bar{x} < 0$, shaded) spans $\alpha \in [0.26, 0.45]$, with lower crossing matching the theoretical prediction within 5%.

Table 4: Mechanism Discrimination Probes at $C = 100$ ($N = 1$ deterministic run; source: `results/mechanism_probes/`).

Probe		α	$\bar{x}$	Dwell
P1: β_2 **Sweep** ($\alpha = 0.8$ fixed)	$\beta_2 = 0.900$	0.80	$+0.376$	0.62
	$\beta_2 = 0.990$	0.80	$+0.170$	0.50
	$\beta_2 \geq 0.999$	0.80	$+0.173$	0.50
P2: α **Sweep** (β_2 fixed)	$\beta_2 = 0.900$	all α	> 0	> 0.60
	$\beta_2 = 0.999$	0.30	-0.180	0.19
	$\beta_2 = 0.999$	0.20	$+0.205$	0.29
	$\beta_2 = 0.999$	0.80	$+0.173$	0.50

in this regime requires sufficiently long non-burst streaks ($g_t = -1$) to push x_t rightward past zero. However, diagnostic analysis ($N = 500$ seeds, source: `results/mechanism_probes/gap_tail_diagnostic.csv`) reveals that across $T = 550$ steps, virtually all seeds experience a maximum negative-gradient run of 40–45 steps (well above the ≈ 25 steps needed to traverse the interval). Consequently, the correlation between maximum negative streak length and failure is near zero ($\text{Corr}(k_{\text{neg, max}}, \text{failed}) = 0.016$). Rather than being driven by the rarity of gap tail events, terminal seed positions at small C are dominated by diffusion and the arrival timing of the final few bursts before T.

4.6 Data Collapse Quality

Against the memory ratio $\rho = \frac{\tau_2}{T_{\text{burst}}}$, the interpolated 0.5-crossing ρ_{50} is defined only for $C \in \{10, 30, 100\}$ where the transition lies inside the probed range; it spans 0.51 decades. For $C \geq 300$, all probed cells exceed 0.5, so a universal single-curve collapse claim must be withdrawn. The transition is governed primarily by C (large-C two-regime structure) rather than ρ alone.

5 Discussion, Practical Implications & Limitations

5.1 Two-Regime Summary

Our mechanism discrimination experiments reveal two structurally distinct divergence regimes at large gradient burst spacing:

1. **Regime I (v-Collapse, $\tau_2 \lesssim T_{\text{burst}}$):** The second-moment estimate decays between bursts, amplifying the effective learning rate on non-burst steps. This regime fails unconditionally for all learning rates α tested (Probe P2, $\beta_2 = 0.9$ column, $C = 100$). The fix (AMSGrad) is precisely targeted at this regime.

2. **Regime II (Post-Burst Drift, $\tau_2 \gg T_{\text{burst}}$):** With high β_2, v_t remains approximately constant. Failure is governed by the traverse condition $\alpha(C - s^*(C)) \gtrsim 2\sqrt{C}$, where $s^*(C) = \tau_1 \ln((1-\beta_1)C + 1)$ is the protected window during which the first moment $\tilde{m}_t$ remains positive after a burst, and $\tau_1 = 1/(1-\beta_1)$ is the first-moment memory horizon. At $C = 100$, the predicted threshold $\alpha^* = 0.263$ matches the empirically bisected boundary $\alpha^* = 0.251$ to within 5%.

5.2 Practical Conservatism of AMSGrad

AMSGrad successfully resolves Regime I by enforcing $\hat{v}_t \geq \hat{v}_{t-1}$. This guarantees $\Gamma_t \succeq 0$ and restores sub-linear regret $\mathcal{O}(\sqrt{T})$ under decreasing step sizes. However, in non-stationary deep learning settings, the historical maximum $\hat{v}_t$ can be locked by a single early outlier gradient burst, depressing future coordinate-wise learning rates and inducing practical sluggishness.

5.3 Regret Exponents Under Constant Step Size

Table 5 reports empirical windowed log-log regret slopes λ (fit over $t \in [0.1T, T]$) for all three optimizers on the stochastic benchmark ($C = 20, \delta = 0.05$, constant $\alpha = 0.8$, $N = 30$ seeds; source: `results/mechanism_probes/lambda_regret_slopes.json`).

Table 5: Empirical windowed log-log regret slopes λ under constant step size ($\alpha = 0.8$). *Caveat:* these are finite-time empirical descriptors only; see text.

Optimizer	λ	Transient Regime
Adam ($\beta_2 = 0.5$)	1.2275	Positive divergence ($\mathbb{E}[x_T] > 0$)
AMSGrad (raw)	1.3069	Finite-window offset artifact
AdaGrad	1.4197	Finite-window offset artifact

Reconciliation: Apparent Super-Linear Slopes vs. $\mathcal{O}(T)$ Asymptotic Bounds. On any compact domain $\mathcal{F} = [-1, 1]$ with bounded subgradients ($\mathbb{E}[g_t] = \delta$), the instantaneous expected regret is $\mathbb{E}[g_t(x_t - x^*)] \leq 2\delta$, strictly capping cumulative regret at $\mathbb{E}[R_T] \leq 2\delta T = \mathcal{O}(T)$. The apparent slopes $\lambda > 1$ in Table 5 are finite-window fitting artifacts:

1. **Initialization Offset:** Because parameters initialize at $x_1 = 0$ near the optimal basin $x^* = -1$, early instantaneous regret is near zero (or temporarily negative due to sample noise). Fitting a pure power law $R(t) \propto t^\lambda$ to an affine trajectory $R(t) \approx c(t - t_0)$ with a positive delay $t_0 > 0$ yields an apparent elasticity $\frac{d \ln R}{d \ln t} = \frac{t}{t - t_0} > 1$.
2. **Constant Step Size vs. Theoretical Convergence:** Under fixed $\alpha = 0.8$, step sizes do not decay ($\alpha_t \not\to 0$). With effective drift $\mu \approx -20$ and noise scale $\sigma_{\text{eff}} \approx 13$ ($\hat{v}_t \approx 400$), AMSGrad maintains a persistent stationary distribution across $[-1, 1]$ with mean ≈ -0.08, generating asymptotically linear regret $\Theta(T)$. The optimal sublinear bound $\mathcal{O}(\sqrt{T})$ strictly requires $\alpha_t \propto 1/\sqrt{t}$, under which AMSGrad concentrates at $x^* = -1$ (Figure 3).

Therefore, the λ values in Table 5 describe transient log-log curvature under constant step size and must *not* be interpreted as asymptotic convergence exponents or theoretical regret rates.

5.4 Conclusion

In this work, we presented a complete, reproducible replication of the Adam $\to$ AMSGrad transition and a pre-registered parametric extension. Our primary findings are:

1. **Exact Counterexample Reproduction:** Both deterministic and stochastic Reddi et al. counterexamples are reproduced, with live Γ_t tracking visually confirming the OCO semi-definiteness breakdown.
2. **Boundary Disappearance:** Under rigorous cycle-averaged metrics, the divergence boundary exists only at $C = 30$ ($k^* = 0.300 \pm 5 \times 10^{-5}$, $\beta_2^* = 0.990 \pm 2 \times 10^{-6}$; bisection half-intervals). For $C \geq 100$, Adam fails unconditionally across all $\beta_2 \in [0.05,\ 1 - 10^{-6}]$, refuting the pre-registered C^{-1} scaling law. At $C = 10$, behavior is mixed: survival at high β_2 and failure at $\beta_2 = 0.5$.
3. **Two-Regime Mechanism:** v-collapse drives failure at low τ_2; post-burst first-moment drift governs failure at large τ_2. The traverse threshold $\alpha^*(C) = 2\sqrt{C}/(C - s^*(C))$ is quantitatively validated at $C = 100$ (5% error) and $C = 300$ (6% error).

Data and Code Availability. All experimental scripts, result files (`results/*.csv`, `results/mechanism_probes/*`), and a reproducible virtual environment specification are archived in the project repository. The repository includes: `experiments/reanalyze_all.py`, `experiments/mechanism_validation.py`, `experiments/make_alpha_star_fig.py`, and the `src/` package. Specific CSV source files are cited inline throughout §4. A public release URL will be added upon acceptance; the full archive can be requested from the authors.

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